

Jiadong Xu

xujiadong24@mailsucas.ac.cn

Personal Profile

Master's student in Electronic Information at the Institute of Software, Chinese Academy of Sciences, with a B.Eng. in Artificial Intelligence from Chongqing University and an MSc in Data Science with Distinction from the University of Bristol. My academic background spans intelligent software development, quality assurance, machine learning, and data science. Beyond a strong interest in embodied AI, I have been actively involved in multiple research projects in robotics, with work contributing to papers accepted by top-tier international conferences.

Education

University of Chinese Academy of Sciences (UCAS) - Institute of Software

M.Eng. Electronic Information

Beijing, China

Sep 2024 - Present

- **Research Focus:** Intelligent Software Development Methods and Quality Assurance (Automated Code Review, Automated Code Repair).

University of Bristol

MSc Data Science with Distinction

Bristol, UK

Sep 2023 - 2024

- **Courses:** Large-Scale Data Engineering, Statistical Computing and Empirical Methods, Introduction to AI and Text Analytics, Visual Analytics.

Chongqing University

B.Eng. Artificial Intelligence

Chongqing, China

Sep 2020 - 2024

- **Courses:** Object-Oriented Programming, Discrete Mathematics, Operating Systems, Computer Networks, Data Structures and Algorithms, Introduction to AI, Machine Learning, Deep Learning, Multivariate Statistical Analysis, Computer Vision, Natural Language Processing.

Work/Research Experience

RealTwin-RL: Bridging Digital Twin Reinforcement Learning to Real-World VLA

Jiadong Xu, Yiwen Wang, Xiaoqi Li, Zhengyong Zhu, Zhanyang Gu, Hongwei Fan, Quanxin Hu, Haomin

2026

Wu, Zhanying Shao, Ran Cheng, Jiafeng Xu, Li Yang, Hao Dong

Proposed a Real2Sim2Real framework for VLA reinforcement learning post-training in digital twins, using semantic milestones to estimate task progress and construct dense potential-based rewards.

Introduced a milestone-conditioned reality discriminator to calibrate rewards and suppress exploitation of simulation artifacts, with validation on five real-world tasks across two embodiments.

(Submitted to ICRA 2027) arXiv submission forthcoming.

IntentReviewer: Learning Review Intents for Code Review Comment Generation

First-author manuscript under double-anonymous review

2026

Research on automated code review.

AnchorVLA4D: an Anchor-Based Spatial-Temporal Vision-Language-Action Model for Robotic Manipulation

Juan Zhu, Zhanying Shao, Xiaoqi Li, Ethan Morgan, Jiadong Xu, Hongwei Fan, Hao Dong

2026

Proposed a spatial-temporal Vision-Language-Action framework that preserves initial scene context through anchor images and improves robotic manipulation under occlusion and spatial ambiguity.

Incorporated a lightweight spatial encoder to enhance geometric reasoning with minimal inference overhead, improving both benchmark and real-world manipulation performance.

(Under Submission) Preprint: arXiv:2603.12730

AT-VLA: Adaptive Tactile Injection for Enhanced Feedback Reaction in Vision-Language-Action Models

Xiaoqi Li, Muhe Cai, Jiadong Xu, Juan Zhu, Hongwei Fan, Yan Shen, Guanghui Ren, Hao Dong

2025

Proposed an adaptive tactile injection mechanism that dynamically determines when and where tactile feedback should be incorporated into a Vision-Language-Action model, reducing interference with pretrained capabilities.

Introduced a dual-stream tactile reaction design to support rapid closed-loop responses for contact-rich robotic manipulation.

(Accepted by CVPR 2026)

Imagine2Act: Leveraging Object-Action Motion Consistency from Imagined Goals for Robotic Manipulation

Liang Heng*, Jiadong Xu*, Yiwen Wang*, Xiaoqi Li*, Muhe Cai, Yan Shen, Juan Zhu, Guanghui Ren, Hao Dong

2025

Designed a robust imagined goal point cloud generation module to provide zero-shot semantic and geometric constraints, enhancing geometric awareness for high-precision tasks.
Developed an Object-Action Consistency Learning strategy with soft pose supervision to explicitly align predicted end-effector motion with generated object transformation.

(Accepted by ICRA 2026) Preprint: arXiv:2509.17125

MLA: A Multisensory Language-Action Model for Multimodal Understanding and Forecasting in Robotic Manipulation

Zhuoyang Liu*, Jiaming Liu*, Jiadong Xu, Nuowei Han, Chenyang Gu, Hao Chen, Kaichen Zhou, Renrui Zhang, Kai Chin Hsieh, Kun Wu, Zhengping Che, Jian Tang, Shanghang Zhang

2025

Proposed an encoder-free multimodal alignment mechanism, repurposing the LLM backbone to directly integrate and interpret 2D vision, 3D point clouds, and tactile cues.
Introduced a future multisensory generation post-training strategy to enhance physical dynamics modeling by jointly forecasting future states of image, point cloud, and tactile modalities.

(Accepted by ICRA 2026) Preprint: arXiv:2509.26642

Data Science Project

Bristol, UK

May 2024 – Sep 2024

Investigated the effectiveness of time-series data in enhancing the recognition accuracy of robot sensors, specifically applied in a robotics context.

Undergraduate Thesis

Chongqing, China

Nov 2023 – May 2024

Research focused on improving automated code review comment generation using Reinforcement Learning from Human Feedback (RLHF).

Students Research Training Program

Chongqing, China

May 2021 - May 2022

Attempted to enhance the recognition and segmentation of camera features in video analysis models. Designed feature-enhanced fusion modules that leveraged Self-Attention and Guided-Attention mechanisms to construct modular Co-Attention Networks (MCAN).

Research Interests

Embodied Intelligence	Building autonomous systems that perceive, reason, and act in unstructured physical environments, with a focus on generalization across tasks and scenes.
Embodied Reinforcement Learning	Learning robotic manipulation policies through interaction, with a focus on reward design, VLA post-training, and real-world transfer.
Robot Learning and Action Models	Studying how robots understand tasks, model environments, and generate executable actions for manipulation.
Multimodal Robot Perception	Building scene representations from text, images, point clouds, and tactile inputs to support robotic task understanding and manipulation decisions.
Simulation for Real-World Robot Learning	Using physics-based simulators, digital twins, and learned world models to improve policy learning, evaluation, and generalization in real-world robotic manipulation.

Skills

Programming	Python, C/C++, HTML, SQL.
Miscellaneous	Linux, Microsoft Office, Markdown, Git.
Soft Skills	Time Management, Teamwork, Problem-solving, Communication, Documentation.
GRE	Verbal 152, Quantitative 168, Writing 3.0.

Languages

English	IELTS Academic Test: 6.5 overall with reading at 8.0, listening at 6.5, writing and speaking at 6.0. College English Test Band 6(CET6): 549.
Chinese	Native proficiency.